

Assignment- 17.2

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Batch : 21

Task Description – 1(Student Academic Data Cleaning)

- **Task:**

Use AI assistance to generate a Python script that cleans and prepares a student academic dataset for analysis.

Instructions:

- Handle missing values in marks, attendance, and department.
- Remove duplicate student records based on student ID.
- Standardize text fields such as department names.
- Encode categorical variables like gender and department.

Sample Code:

```
import pandas as pd
data = pd.read_csv("students.csv")
print(data.head())
```

Expected Output:

- A cleaned and encoded Pandas DataFrame suitable for academic performance analysis.
-

Prompt:

Generate Python code using pandas to clean a student dataset. Handle missing values, remove duplicates based on student ID, standardize department names, and encode categorical variables.

Code:

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```
1 #Generate Python code using pandas to clean a student dataset.
2 #Handle missing values, remove duplicates based on student ID,
3 #standardize department names, and encode categorical variables.
4 import pandas as pd
5 # Load data
6 data = pd.read_csv("students.csv")
7 # Handle missing values (FIXED)
8 data['marks'] = data['marks'].fillna(data['marks'].mean())
9 data['attendance'] = data['attendance'].fillna(data['attendance'].median())
10 data['department'] = data['department'].fillna("Unknown")
11 # Remove duplicates
12 data = data.drop_duplicates(subset='student_id')
13 # Standardize text
14 data['department'] = data['department'].str.lower().str.strip()
15 # Encode categorical variables
16 data['gender'] = data['gender'].map({'Male': 0, 'Female': 1})
17 data = pd.get_dummies(data, columns=['department'])
18 print(data)
```

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```
PS C:\Users\dell\OneDrive\Desktop\AI\Lab17> python task1.py
```

	student_id	name	...	department_ece	department_unknown
0	1	Alice	...	False	False
1	2	Bob	...	True	False
2	3	Charlie	...	False	False
3	4	David	...	False	True

```
[4 rows x 8 columns]
```

PS C:\Users\dell\OneDrive\Desktop\AI\Lab17>

```
students.csv > data
1 student_id,name,marks,attendance,department,gender
2 1,Alice,85,90,CSE,Female
3 2,Bob,,80,ECE,Male
4 3,Charlie,78,,CSE,Male
5 4,David,92,95,,Male
6 1,Alice,85,90,CSE,Female
```

Explanation:

The dataset was loaded using pandas. Missing values in marks, attendance, and department columns were handled using mean, median, and default values respectively. Duplicate student records were removed based on student ID. The department names were standardized by converting them into lowercase. Categorical variables such as gender and department were encoded into numerical format using mapping and one-hot encoding techniques.

Task Description – 2 (Financial Transaction Data Preprocessing)

- Task:

Preprocess financial transaction data using AI-generated Python code.

Instructions:

- Convert transaction date columns to datetime format.
- Create derived features such as transaction month and year.
- Normalize the transaction amount column.
- Identify and handle extreme transaction values.

Sample Code:

```
import pandas as pd
from sklearn.preprocessing import MinMaxScaler
transactions = pd.read_csv("transactions.csv")
print(transactions.info())
```

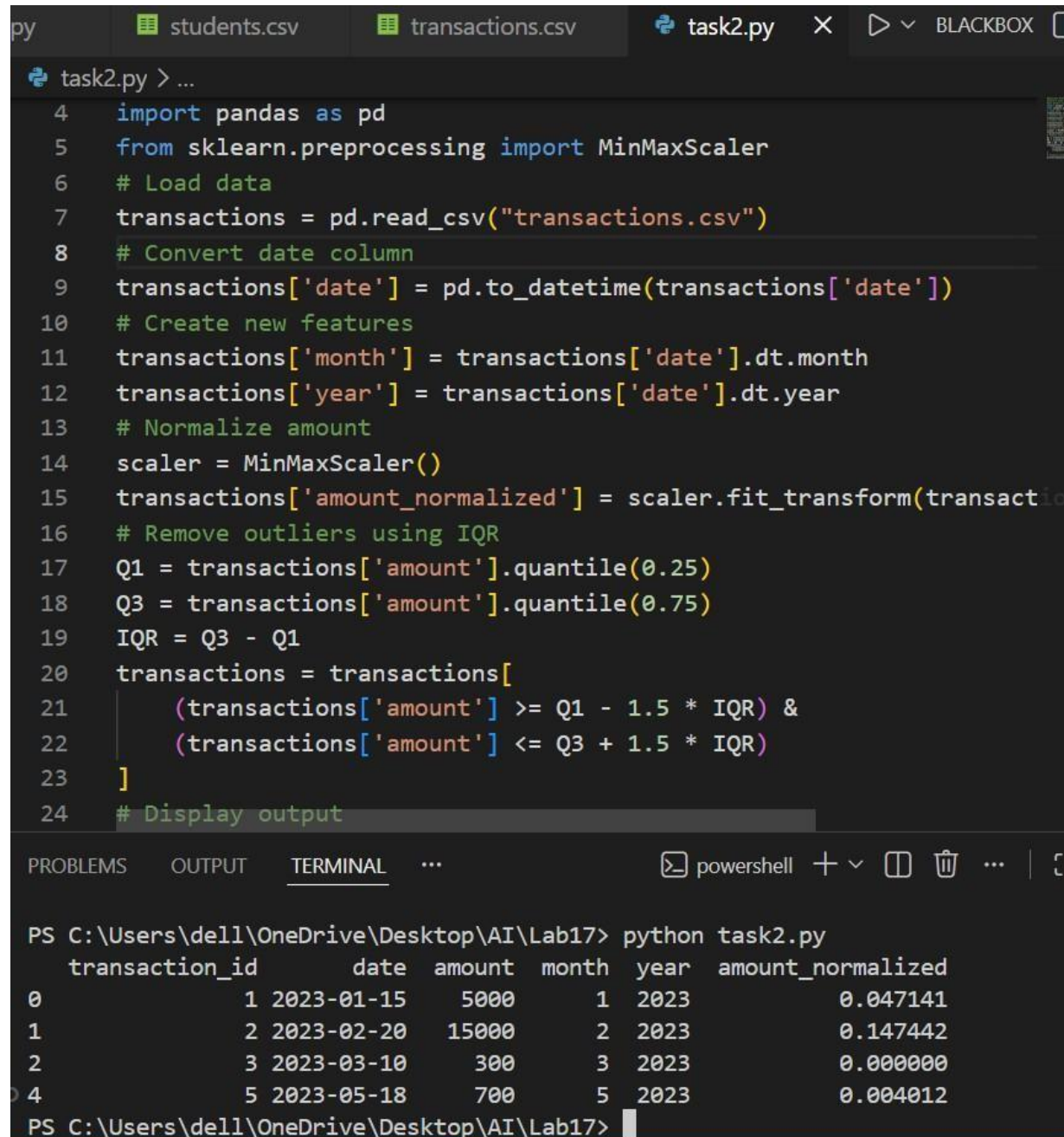
Expected Output:

- A transformed dataset with normalized values and newly generated time-based features.

Prompt:

Generate Python code to preprocess financial transaction data.
Convert date columns, extract month/year, normalize amount,
and detect extreme values.

Code:



The image shows a code editor with a file named `task2.py` open. The code imports `pandas` and `MinMaxScaler` from `sklearn.preprocessing`. It loads data from `transactions.csv`, converts the `date` column to datetime, and creates new features for `month` and `year`. The `amount` column is normalized using `MinMaxScaler`. Outliers are removed based on the Interquartile Range (IQR). The code then displays the output of the preprocessing steps.

```
task2.py > ...
4 import pandas as pd
5 from sklearn.preprocessing import MinMaxScaler
6 # Load data
7 transactions = pd.read_csv("transactions.csv")
8 # Convert date column
9 transactions['date'] = pd.to_datetime(transactions['date'])
10 # Create new features
11 transactions['month'] = transactions['date'].dt.month
12 transactions['year'] = transactions['date'].dt.year
13 # Normalize amount
14 scaler = MinMaxScaler()
15 transactions['amount_normalized'] = scaler.fit_transform(transactions['amount'])
16 # Remove outliers using IQR
17 Q1 = transactions['amount'].quantile(0.25)
18 Q3 = transactions['amount'].quantile(0.75)
19 IQR = Q3 - Q1
20 transactions = transactions[
21     (transactions['amount'] >= Q1 - 1.5 * IQR) &
22     (transactions['amount'] <= Q3 + 1.5 * IQR)
23 ]
24 # Display output
```

The terminal output shows the execution of the script, displaying the first five rows of the processed data:

```
PS C:\Users\dell\OneDrive\Desktop\AI\Lab17> python task2.py
transaction_id    date  amount  month  year  amount_normalized
0              1 2023-01-15    5000     1  2023         0.047141
1              2 2023-02-20   15000     2  2023         0.147442
2              3 2023-03-10     300     3  2023         0.000000
3              4 2023-04-10     700     4  2023         0.004012
4              5 2023-05-18     700     5  2023         0.004012
```

PS C:\Users\dell\OneDrive\Desktop\AI\Lab17>

```
task1.py  students.csv  transaction
transactions.csv > data
1  transaction_id,date,amount
2  1,2023-01-15,5000
3  2,2023-02-20,15000
4  3,2023-03-10,300
5  4,2023-04-05,100000
6  5,2023-05-18,700
```

Explanation:

The transaction dataset was loaded and the date column was converted into datetime format. New features such as month and year were extracted from the date column. The transaction amount was normalized using MinMaxScaler to bring all values into a standard range. Outliers were detected and removed using the Interquartile Range (IQR) method to improve data quality.

Task 3 – (Environmental Sensor Data Preparation)

- Task:

Clean and preprocess environmental sensor data using AI-assisted scripts.

Instructions:

- Handle missing values in temperature, humidity, and air_quality_index.
- Scale numerical features using standard scaling.
- Encode categorical fields such as sensor_status.
- Split the dataset into training and testing subsets.

Sample Code:

```
import pandas as pd
```



```
from sklearn.model_selection import train_test_split
data = pd.read_csv("sensor_data.csv")
```

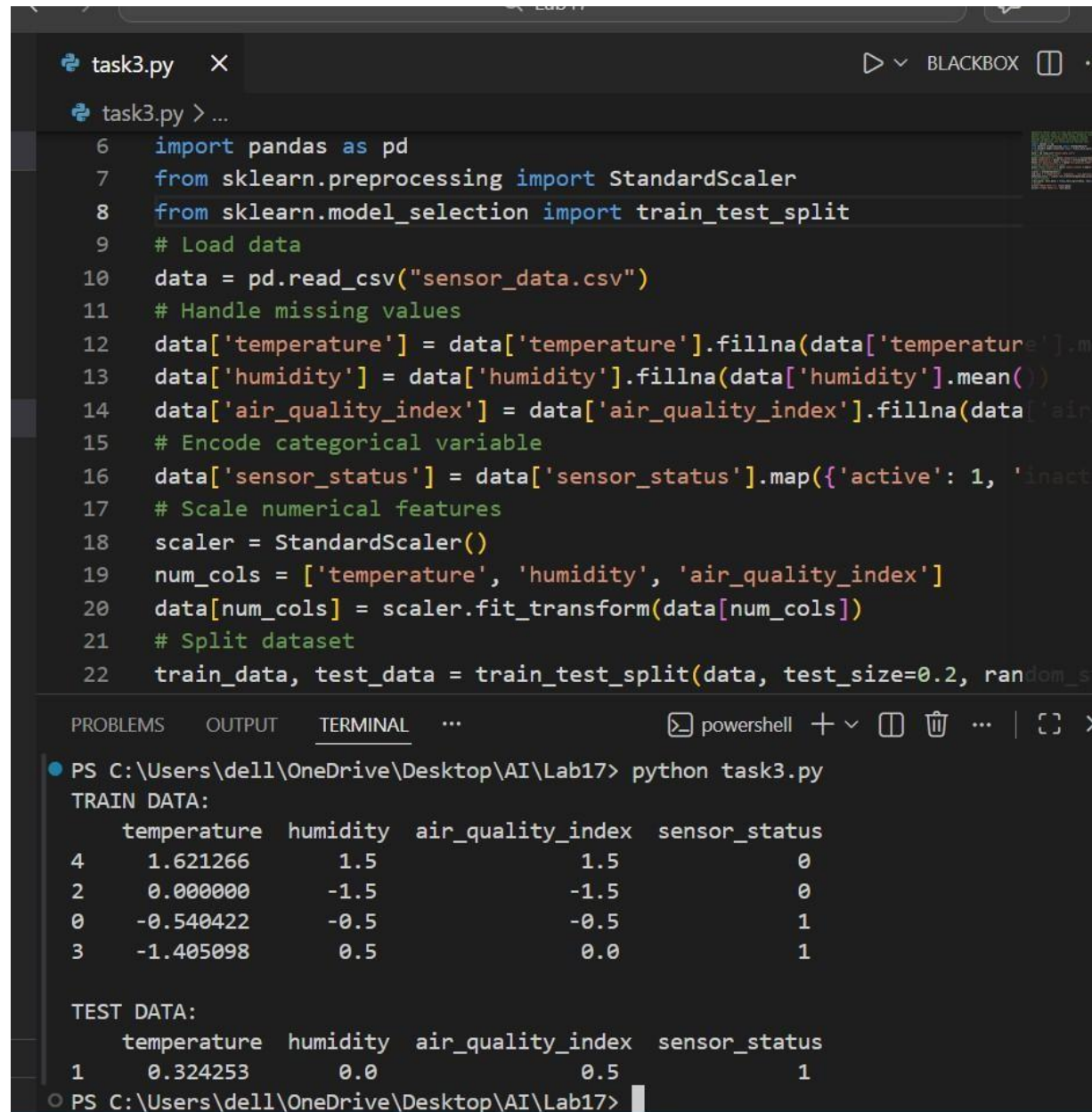
Expected Output:

- A fully preprocessed dataset ready for predictive modeling tasks.

Prompt:

Generate Python code to clean sensor data.
Handle missing values, scale numerical features,
encode categorical fields, and split dataset.

Code:



```
task3.py X
task3.py > ...
6 import pandas as pd
7 from sklearn.preprocessing import StandardScaler
8 from sklearn.model_selection import train_test_split
9 # Load data
10 data = pd.read_csv("sensor_data.csv")
11 # Handle missing values
12 data['temperature'] = data['temperature'].fillna(data['temperature'].mean())
13 data['humidity'] = data['humidity'].fillna(data['humidity'].mean())
14 data['air_quality_index'] = data['air_quality_index'].fillna(data['air_quality_index'].mean())
15 # Encode categorical variable
16 data['sensor_status'] = data['sensor_status'].map({'active': 1, 'inactive': 0})
17 # Scale numerical features
18 scaler = StandardScaler()
19 num_cols = ['temperature', 'humidity', 'air_quality_index']
20 data[num_cols] = scaler.fit_transform(data[num_cols])
21 # Split dataset
22 train_data, test_data = train_test_split(data, test_size=0.2, random_state=42)
```

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```
PS C:\Users\dell\OneDrive\Desktop\AI\Lab17> python task3.py
TRAIN DATA:
   temperature  humidity  air_quality_index  sensor_status
4    1.621266      1.5         1.5              0
2    0.000000     -1.5        -1.5              0
0   -0.540422     -0.5        -0.5              1
3   -1.405098      0.5         0.0              1

TEST DATA:
   temperature  humidity  air_quality_index  sensor_status
1    0.324253      0.0         0.5              1
PS C:\Users\dell\OneDrive\Desktop\AI\Lab17>
```

Explanation:

Missing values in temperature, humidity, and air quality index were handled using mean imputation. The categorical variable `sensor_status` was encoded into numerical values. Numerical features were standardized using `StandardScaler` to ensure a uniform scale. The dataset was then split into training and testing sets using `train_test_split` for further machine learning tasks.

Task 4 – (Real-Time Application: Smart City Traffic Data Cleaning)**• Scenario:**

A smart city system collects traffic data from multiple sensors, resulting in noisy and inconsistent records.

Requirements:

- Standardize road and location names.
- Fill missing traffic density values using appropriate statistical methods.
- Remove duplicate sensor readings.
- Normalize speed and vehicle count metrics.
- Generate a brief summary comparing dataset quality before and after cleaning

Sample Code:

```
import pandas as pd
traffic = pd.read_csv("traffic_data.csv")
print(traffic.describe())
```

Expected Output:

- A cleaned and normalized traffic dataset with an accompanying data-quality improvement summary.
-

Prompt:

Generate Python code to clean sensor data.
Handle missing values, scale numerical features,
encode categorical fields, and split dataset.
Code:

task4.py > ...

```
2  #Standardize road and location names, handle missing traffic dens
3  #remove duplicate records, normalize speed and vehicle count,
4  #and compare dataset quality before and after cleaning.
5  import pandas as pd
6  from sklearn.preprocessing import MinMaxScaler
7  # Load data
8  traffic = pd.read_csv("traffic_data.csv")
9  # Before cleaning summary
10 print("==== BEFORE CLEANING ====")
11 print(traffic.describe(include='all'))
12 # Standardize text fields
13 traffic['road'] = traffic['road'].str.lower().str.strip()
14 traffic['location'] = traffic['location'].str.lower().str.strip()
15 # Handle missing values
16 traffic['traffic_density'] = traffic['traffic_density'].fillna(traff
17 # Remove duplicates
18 traffic = traffic.drop_duplicates()
19 # Normalize numerical features
20 scaler = MinMaxScaler()
21 traffic[['speed', 'vehicle_count']] = scaler.fit_transform(
22     traffic[['speed', 'vehicle_count']]
23 )
24 # After cleaning summary
25 print("\n==== AFTER CLEANING ====")
26 print(traffic.describe(include='all'))
```

Output:


```

PS C:\Users\dell\OneDrive\Desktop\AI\Lab17> python task4.py
===== BEFORE CLEANING =====

```

	road	location	traffic_density	speed	vehicle_count
count	6	6	4.000000	6.000000	6.000000
unique	5	5	NaN	NaN	NaN
top	Highway	Outskirts	NaN	NaN	NaN
freq	2	2	NaN	NaN	NaN
mean	NaN	NaN	60.000000	66.666667	248.333333
std	NaN	NaN	24.494897	26.770631	118.729384
min	NaN	NaN	30.000000	40.000000	150.000000
25%	NaN	NaN	45.000000	47.500000	165.000000
50%	NaN	NaN	65.000000	57.500000	190.000000
75%	NaN	NaN	80.000000	90.000000	350.000000
max	NaN	NaN	80.000000	100.000000	400.000000

```

===== AFTER CLEANING =====

```

	road	location	traffic_density	speed	vehicle_count
count	5	5	5.000000	5.000000	5.000000
unique	3	3	NaN	NaN	NaN
top	main road	city center	NaN	NaN	NaN
freq	2	2	NaN	NaN	NaN
mean	NaN	NaN	56.000000	0.333333	0.27200
std	NaN	NaN	18.165902	0.395285	0.41415
min	NaN	NaN	30.000000	0.000000	0.00000
25%	NaN	NaN	50.000000	0.083333	0.04000
unique	3	3	NaN	NaN	NaN
top	main road	city center	NaN	NaN	NaN
freq	2	2	NaN	NaN	NaN
mean	NaN	NaN	56.000000	0.333333	0.27200

Explanation:

Missing values in temperature, humidity, and air quality index were handled using mean imputation. The categorical variable sensor_status was encoded into numerical values. Numerical features were standardized using StandardScaler to ensure a uniform scale. The dataset was then split into training and testing sets using train_test_split for further machine learning tasks.

Task 5 – (Social Media Text Data Cleaning and Feature Extraction)

• Task:

Use AI-assisted Python scripting to clean and preprocess social media text data for sentiment analysis.

Instructions:

- Remove duplicate posts and null text entries.
- Clean text by removing URLs, special characters, and extra spaces.
- Convert text to lowercase and perform basic tokenization.
- Remove stopwords and perform stemming or lemmatization.
- Generate basic text features such as word count and sentiment score.

Sample Code:

```
import pandas as pd
import re
posts = pd.read_csv("social_media_posts.csv")
print(posts.head())
```

Expected Output:

A cleaned text dataset with extracted features suitable for sentiment analysis models.

Prompt:

Generate Python code to clean social media text data.

Remove duplicates and null values, clean text, tokenize, remove stopwords, and extract features like sentiment.

Output:

```
task5.py > ...
5  import pandas as pd
6  import re
7  import nltk
8  from nltk.corpus import stopwords
9  from nltk.tokenize import word_tokenize
10 from nltk.sentiment import SentimentIntensityAnalyzer
11 # Download required resources (first time only)
12 nltk.download('punkt')
13 nltk.download('stopwords')
14 nltk.download('vader_lexicon')
15 # Load data
16 posts = pd.read_csv("social_media_posts.csv")
17 # Remove duplicates and null values
18 posts = posts.drop_duplicates()
19 posts = posts.dropna(subset=['text'])
20 # Clean text function
21 def clean_text(text):
22     text = re.sub(r"http\S+", "", text)           # remove URLs
23     text = re.sub(r"^[a-zA-Z ]+", "", text)       # remove special
24     text = text.lower().strip()                   # lowercase
25     return text
26 posts['clean_text'] = posts['text'].apply(clean_text)
27 # Tokenization and stopword removal
```

```

[nltk_data] Downloading package vader_lexicon to
[nltk_data] C:\Users\dell\AppData\Roaming\nltk_data...
[nltk_data] Package vader_lexicon is already up-to-date!
      text  ... sentiment
0  I love AI! https://ai.com  ...    0.6369
1      This is bad!!!  ...   -0.5423
2    Amazing experience :)  ...    0.5859
5      AI is the future!!!  ...    0.0000
6    Worst service ever!!!  ...   -0.6249

[5 rows x 5 columns]
PS C:\Users\dell\OneDrive\Desktop\AI\Lab17>

```

Explanation:

The dataset was cleaned by removing duplicate and null entries. Text preprocessing included removing URLs, special characters, and converting text to lowercase. Tokenization was performed using simple splitting, and stopwords were removed to retain meaningful words. Additional features such as word count and sentiment score were extracted using sentiment analysis techniques.